Phishing Detection & Awareness Report
Email Threat Analysis

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Cybersecurity Internship Project Report

1. Introduction

Phishing is one of the most common and dangerous cyber threats facing individuals and organizations today. It involves attackers sending deceptive emails, messages, or websites that are designed to look trustworthy in order to trick people into giving away sensitive information — such as passwords, bank account details, or personal data.

This report presents the findings of a phishing email detection exercise carried out as part of a cybersecurity internship project. Using real-world phishing email samples, the report identifies attack patterns, explains how each deceptive technique works in plain language, and provides practical guidance to help employees recognize and avoid phishing attempts.

The goal is not to overwhelm the reader with technical jargon, but to build a clear, simple understanding of what phishing looks like, why it works, and what you can do to stay safe.

2. Objective

This report aims to achieve the following goals:

- Identify and document real-world phishing email samples collected during the internship period.
- Analyze each email to highlight specific elements that make it suspicious or dangerous.
- Classify the risk level of each phishing attempt — Low, Medium, or High — based on potential harm.
- Explain, in simple terms, how each type of phishing attack attempts to deceive the recipient.
- Provide actionable prevention tips and best practices that any employee can follow.
- Help build a security-aware culture within the organization by making cybersecurity knowledge accessible.

3. Methodology

The following steps were followed to complete this analysis:

Step 1 — Sample Collection

Phishing email samples were gathered from publicly available security research databases, controlled honeypot inboxes, and curated phishing repositories maintained by cybersecurity organizations. All samples used in this report are either anonymized or sourced from public domain archives.

Step 2 — Header & Metadata Review

Each email's sender address, reply-to field, routing headers, and domain information were examined to detect spoofing, domain impersonation, or mismatched routing paths — all common signs of a forged email.

Step 3 — Content Analysis

The email body was reviewed for psychological manipulation tactics such as urgency, fear, reward promises, and impersonation of trusted entities (banks, tech companies, employers). Grammar, spelling, and branding inconsistencies were also noted.

Step 4 — Link & Attachment Inspection

All embedded links and attachments were examined in a secure, isolated environment. Link destinations were checked against known malicious URL databases and analyzed for signs of deception such as misleading domain names, extra subdomains, URL shorteners, or typosquatting.

Step 5 — Risk Classification

Each email was assigned a risk level (Low, Medium, or High) based on the severity of potential harm, the sophistication of the deception, and the likelihood of a typical user falling victim to the attack.

Step 6 — Documentation & Recommendations

All findings were compiled into this structured report along with plain-language explanations and prevention guidelines designed for non-technical readers.

4. Email Analysis

The following section presents an in-depth analysis of five phishing email samples. Each sample is examined for deceptive tactics, suspicious links, risk level, and attack explanation.

Email Sample #1 — Analysis	
Subject Line	*URGENT: Your Account Has Been Compromised — Verify Now*

Email Content	`From: security-alert@paypa1-support.com`
	`To: customer@youremail.com`
	
	`Dear Valued Customer,`
	
	`We have detected suspicious login activity on your account.`
	`Your account has been TEMPORARILY SUSPENDED for your safety.`
	
	`To restore access, you must verify your identity within`
	`24 hours or your account will be permanently closed.`
	
	`>> Click Here to Verify Your Account Now <<`
	`http://paypa1-support-verify.xyz/login`
	
	`Thank you for your prompt attention.`
	`PayPal Security Team`

Phishing Indicators	• Sender domain is 'paypa1-support.com' — uses the number '1' instead of the letter 'l' to mimic PayPal. This is called typosquatting. • Creates extreme urgency: '24 hours or your account will be permanently closed' — designed to stop you thinking clearly. • The link destination (.xyz domain) has no connection to the real PayPal website (paypal.com). • Generic greeting 'Dear Valued Customer' — legitimate companies use your actual name. • Excessive use of capital letters and exclamation to trigger panic. • No official PayPal branding, case number, or verifiable contact information.
Link Analysis	**`Suspicious URL: http://paypa1-support-verify.xyz/login`** The URL uses 'paypa1' (with the number 1) instead of 'paypal' and uses a .xyz top-level domain — a common trick used to create fake websites that look real at first glance. Clicking this link would take you to a fake login page designed to steal your username and password.
Risk Level	**● HIGH RISK**
How This Attack Works	This is a classic 'Account Takeover' phishing attack. The attacker impersonates PayPal and creates a fake emergency to frighten the victim into clicking a link and entering their login credentials. Once the victim submits their details on the fake website, the attacker captures them and gains full access to the real account. This can lead to financial loss and identity theft.

Email Sample #2 — Analysis

Subject Line	*Action Required: Invoice #8821 Payment Due — Please Review*

Email Content	`From: accounts@supp1ier-invoices.com`
	`To: finance@yourcompany.com`
	
	`Dear Finance Team,`
	
	`Please find attached Invoice #8821 for services rendered`
	`in the amount of $12,450.00, due by end of this week.`
	
	`We have recently updated our banking details.`
	`Please use the NEW account information below for payment:`
	
	`Bank: First National Bank`
	`Account Name: Global Supplies Ltd.`
	`Account No: 7834920011`
	`Routing No: 021000021`
	
	`Kindly confirm receipt of this email.`
	`Regards, Accounts Team — Global Supplies Ltd.`

Phishing Indicators	• Sender domain 'supp1ier-invoices.com' uses the number '1' in place of the letter 'l' — mimicking a legitimate supplier domain. • Urgency tactic: 'due by end of this week' pressures quick, unverified action. • The email asks the finance team to update banking details — a classic Business Email Compromise (BEC) red flag. • No prior relationship or purchase order number is referenced to verify legitimacy. • The request to 'confirm receipt' is designed to confirm the email address is active. • No digital signature, contract reference, or verifiable contact phone number provided.
Link Analysis	**`Suspicious URL: accounts@supp1ier-invoices.com (spoofed sender domain)`** There is no malicious link in this email — the attack is carried out entirely through social engineering. The attacker has registered a domain name that closely resembles a real supplier's domain. If the finance team transfers money to the listed account without verifying through an independent channel, the funds are sent directly to the attacker.
Risk Level	**● MEDIUM RISK**
How This Attack Works	This is a 'Business Email Compromise' (BEC) attack targeting the finance department. The attacker impersonates a known supplier and requests a change to payment bank details. If the employee does not independently verify this change by calling the supplier directly using a known phone number, they could accidentally transfer company funds to a fraudster's account. BEC attacks have caused billions of dollars in losses globally.

Email Sample #3 — Analysis

Subject Line	*Your Package Could Not Be Delivered — Schedule Redelivery*

Email Content	`From: delivery-update@dhl-trackingportal.info`
	`To: customer@youremail.com`
	
	`Dear Customer,`
	
	`We attempted to deliver your package today but were unable to complete the delivery.`
	
	`Tracking Number: DHL-48392-XK`
	
	`To schedule a redelivery or collect from our depot, please confirm your delivery address and pay the small redelivery fee of $2.99:`
	
	`Schedule Redelivery: http://dhl-trackingportal.info/redeliver`
	
	`Please respond within 5 days or your parcel will be returned.`
	`DHL Express Customer Service`

Phishing Indicators	• Sender domain 'dhl-trackingportal.info' is not an official DHL domain (official domains end in @dhl.com). • Requests a small payment ($2.99) which is used to collect real card details for larger fraudulent charges. • Vague delivery details — no real DHL tracking number would look like 'DHL-48392-XK' (format is incorrect). • Creates mild urgency: '5 days or parcel returned' — enough pressure to act without verifying. • Generic 'Dear Customer' greeting instead of your name.
Link Analysis	**Suspicious URL: http://dhl-trackingportal.info/redeliver** The domain 'dhl-trackingportal.info' is not affiliated with DHL in any way. Legitimate DHL communications come from @dhl.com. Clicking this link and entering payment information would give the attacker your full card number, expiry date, and CVV — enabling large unauthorized charges far beyond the stated $2.99 fee.
Risk Level	**● LOW RISK**
How This Attack Works	This is a 'Delivery Notification Scam' — a low-cost, high-volume phishing attack that exploits the rise in online shopping. Attackers send this email to thousands of addresses, knowing many recipients will recently have ordered something and may be expecting a delivery. The small fee request feels harmless but is used to capture valid card details. This type of attack is classified as Low risk relative to others, but remains financially harmful.

5. Risk Classification Summary

The table below summarizes the risk classification for each email sample analyzed in this report. Risk levels are assigned based on potential financial harm, data sensitivity targeted, and sophistication of the deception.

Email Sample	Subject	Risk Level	Primary Threat
Sample #1	*Your Account Has Been Compromised*	**HIGH**	Credential Harvesting
Sample #2	*Action Required: Invoice #8821 Due*	**MEDIUM**	Business Email Compromise
Sample #3	*Package Delivery Notification*	**LOW**	Smishing / Link Redirect

HIGH	Immediate threat. Likely to cause significant financial loss, account takeover, or system compromise. Requires urgent user education and technical controls.
MEDIUM	Moderate threat. Can cause internal data loss, unauthorized system access, or business process disruption. Requires verification procedures and user awareness.
LOW	Lower-scale threat. May result in minor financial loss or data exposure. Still requires reporting and awareness. Should not be ignored.

6. How Phishing Works — A Simple Explanation

Phishing is essentially a confidence trick performed online. Here is how a typical phishing attack unfolds, step by step:

Step 1	The Bait — The attacker crafts an email that appears to come from a trusted source: your bank, your employer, a courier company, or a government agency. The message looks convincing and may even include real logos.
Step 2	The Hook — The email contains an urgent call to action. You are told your account is at risk, a payment is overdue, or a prize awaits you. This urgency is designed to make you act quickly without stopping to think.
Step 3	The Trap — You click a link that takes you to a fake website. This site looks identical to a real one but is controlled by the attacker. You enter your login credentials, payment details, or personal information.
Step 4	The Catch — The attacker now has your data. Within minutes, they may log into your real account, drain funds, steal your identity, or sell your information to other criminals.
Step 5	The Cover-Up — Some attackers redirect you to the real website after capturing your data, so you do not immediately realize anything went wrong.

7. Common Phishing Indicators

When reading any email, watch out for these warning signs. The more indicators present, the more suspicious the email should be treated.

- Suspicious sender address — The email comes from an address that does not match the organization it claims to be from (e.g., support@paypa1.com instead of support@paypal.com).
- Urgency or threats — The email tells you that you must act immediately or face negative consequences such as account suspension, missed delivery, or legal action.
- Too-good-to-be-true offers — Promises of prizes, lottery wins, refunds, or large sums of money that you did not expect.
- Generic greetings — Phrases like 'Dear Customer', 'Dear User', or 'Hello' instead of your real name.
- Spelling and grammar mistakes — Professional organizations proofread their communications. Errors can indicate a fraudulent message.
- Mismatched or suspicious links — Hovering over a link shows a web address that does not match the stated destination or the organization's real website.
- Unexpected attachments — Files attached without prior context, especially .zip, .exe, .pdf, or .docx files from unknown senders.
- Requests for sensitive information — Legitimate organizations will never ask for your password, PIN, full banking details, or Social Security Number via email.
- Requests to change payment details — Especially for suppliers or payroll, treat any request to update banking details with extreme caution.

8. Prevention Guidelines

The following guidelines are recommended to reduce the risk of falling victim to phishing attacks at both the individual and organizational level.

For All Employees

- Always verify the sender's email address carefully before opening links or attachments.
- When in doubt, contact the supposed sender directly using a known, verified contact method — not the one provided in the suspicious email.
- Never click links in emails to log into accounts. Instead, open a new browser tab and type the website address directly.
- Enable Multi-Factor Authentication (MFA) on all accounts — this protects you even if your password is stolen.
- Keep your devices and software up to date to protect against malware delivered through phishing links.
- Report suspicious emails to your IT or security team immediately — do not forward or reply to them.

For IT and Security Teams

- Deploy email filtering solutions that automatically detect and quarantine phishing emails.
- Implement DMARC, DKIM, and SPF email authentication protocols to reduce domain spoofing.
- Conduct regular phishing simulation exercises to test employee awareness.
- Maintain a clear, easy-to-use reporting process so employees feel comfortable flagging suspicious emails.
- Enforce the principle of least privilege — limit access to sensitive systems to only those who need it.
- Provide regular cybersecurity awareness training with updated examples of current phishing techniques.

9. Do's and Don'ts for Users

DO ✓	DON'T ✗
✓ Verify the sender's email address carefully before taking any action.	✗ Don't click links in emails to log in — go directly to the website instead.
✓ Hover over links to preview the real URL before clicking.	✗ Don't open unexpected attachments, especially from unknown senders.
✓ Use strong, unique passwords for every account and store them in a password manager.	✗ Don't share your password, PIN, or OTP with anyone — not even IT staff.
✓ Enable Multi-Factor Authentication (MFA) on all important accounts.	✗ Don't act on urgency or threats in emails without independently verifying first.
✓ Report suspicious emails to your IT or security team immediately.	✗ Don't reply to or forward suspected phishing emails.
✓ Keep your operating system and applications updated.	✗ Don't use the phone number or contact details provided in a suspicious email.
✓ When in doubt, call the sender directly using a known number to verify.	✗ Don't dismiss phishing concerns — report even if you are not sure.

10. Conclusion

Phishing remains one of the most effective and widely used forms of cyber attack — not because it exploits complex technical vulnerabilities, but because it exploits human psychology. Attackers rely on trust, fear, urgency, and excitement to cloud our judgment and push us into making hasty decisions.

The five email samples analyzed in this report demonstrate a broad range of phishing tactics, from high-urgency account compromise scams and prize fraud, to sophisticated business email compromise attacks targeting finance teams. While the techniques differ, they all share common elements: impersonation, urgency, deceptive links, and requests for sensitive information.

The good news is that phishing attacks can be effectively countered with awareness, vigilance, and a few simple habits. By learning to recognize the warning signs — suspicious sender addresses, unexpected urgency, mismatched links, requests for sensitive data — employees at every level can serve as the organization's first and most important line of defense.

It is recommended that organizations follow up on this report by implementing regular phishing awareness training, deploying email filtering technologies, and establishing a clear incident reporting procedure. Cybersecurity is not the sole responsibility of the IT department — it is a shared responsibility that requires every person in the organization to stay informed and alert.

Remember

When in doubt — stop, verify, and report.

A moment of caution can prevent a major security incident.